

Ing. Michal Neoral Ph.D.

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EDUCATION

- **Czech Technical University in Prague** [Doctoral degree – Ph.D. \(Philosophiae Doctor\)](#) 2018 – 2025
 - Faculty of Electrical Engineering
 - Informatics / Artificial Intelligence and Biocybernetics
 - thesis topic: Dense Motion Estimation in a Monocular Video
- **Czech Technical University in Prague** [Master's degree in engineering – Ing. \(MSc equivalent\)](#) 2014 – 2017
 - Faculty of Electrical Engineering
 - Open Informatics
 - specialization: Computer Vision and Image Processing
- **Czech Technical University in Prague** [Bachelor's degree in engineering – Bc. \(BSc equivalent\)](#) 2011 – 2014
 - Faculty of Electrical Engineering
 - Cybernetics and Robotics
 - specialization: Robotics

EXPERIENCE

- **Czech Technical University in Prague** [Research Developer](#) Feb 2025 – present
 - Faculty of Electrical Engineering, Center for Machine Perception
 - Detection, classification and fine-grained recognition: FunDive, Local CLS, CEDMO, AISee, Median
- **Czech Technical University in Prague** [Research Scientist](#) Jul 2017 – Jan 2025
 - Computer vision research for Toyota Motor Europe
 - Dense motion estimation in monocular video: optical flow with occlusion reasoning, discovery and segmentation of independently moving objects, long-term dense tracking
 - Basis of the doctoral dissertation and of four patent filings
- **Czech Technical University in Prague** [University Teacher](#) Mar 2018 – present
 - Lab teaching, detailed in the Teaching section below
- **Czech Technical University in Prague** [Researcher and Developer](#) 2012 – Feb 2017
 - Part-time alongside the bachelor's and master's studies: CloPeMa, SyRoTek, Electrolux, Toyota

TEACHING

- **Pattern Recognition and Machine Learning** [Czech Technical University in Prague](#) 2018 – present
 - teaching
 - material preparation
 - development of automatic evaluation of student homeworks
- **Digital Photography Processing** [Czech Technical University in Prague](#) 2020 – 2022
- **Computer Vision Methods** [Czech Technical University in Prague](#) 2018 – 2019

- [C] Jonáš Šerých, Michal Neoral, and Jiří Matas. (2025). **MFTIQ: Multi-Flow Tracker with Independent Matching Quality Estimation**. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, February 2025, Tucson, Arizona USA.
- [C] Michal Neoral, Jonáš Šerých, and Jiří Matas. (2024). **MFT: Long-term Tracking of Every Pixel**. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 6837–6847, January 2024, Waikoloa, Hawaii USA.
- [C] Michal Neoral, Jan Šochman, and Jiří Matas. (2021). **Monocular Arbitrary Moving Object Discovery and Segmentation**. In *The 32nd British Machine Vision Conference*, November 2021, Online.
- [C] Michal Neoral, Jan Šochman, and Jiří Matas. (2019). **Continual Occlusion and Optical Flow Estimation**. Computer Vision – ACCV 2018. Ed. by C.V. Jawahar et al. Cham: Springer International Publishing, 2019, pp. 159–174. ISBN: 978-3-030-20870-7.
- [C] Michal Neoral and Jan Šochman. (2017). **Object Scene Flow with Temporal Consistency**. In *22nd Computer Vision Winter Workshop (CVWW)*. Pattern Recognition and Image Processing Group; TU Wien & PRIP Club; Vienna; Austria, February 2017, ISBN: 978-3-200-04969-7.
- [O] Matej Kristan, Jiří Matas, Pavel Tokmakov, Alan Lukežič, Michael Felsberg, Luka Čehovin Zajc, Khanh-Tung Tran, Xuan-Son Vu, Johanna Björklund, Michal Neoral, Hyung Jin Chang, et al. (2025). **The Third Visual Object Tracking Segmentation VOTS2025 Challenge Results**. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, October 2025, Honolulu, Hawaii USA.
- [O] Adam Schmidt, Mert Asim Karaoglu, Soham Sinha, Mingang Jang, Ho-Gun Ha, Kyungmin Jung, Kyeongmo Gu, Ihsan Ullah, Hyunki Lee, Jonáš Šerých, Michal Neoral, Jiří Matas, et al. (2025). **Point Tracking in Surgery – The 2024 Surgical Tattoos in Infrared (STIR) Challenge**. arXiv preprint arXiv:2503.24306.
- [P] Nikolay Chumerin, Michal Neoral, Jan Šochman, and Jiří Matas (2023). **MONOCULAR-VISION-BASED DETECTION OF MOVING OBJECTS**. EP Patent EP4174770B1, filed Oct 28, 2021, and issued Nov 29, 2023. **Granted Patent**
- [P] Nikolay Chumerin, Jonáš Šerých, Michal Neoral, and Jiří Matas. (2024). **A METHOD AND SYSTEM FOR TRACKING A PIXEL IN A VIDEO**. EP Patent EP4465239A1, filed May 16, 2023, and issued Nov 20, 2024. **Patent Application**
- [P] Nikolay Chumerin, Michal Neoral, Jan Šochman, and Jiří Matas. (2023). **MONOCULAR-VISION-BASED DETECTION OF MOVING OBJECTS**. EP Patent EP4174770A1, filed Oct 28, 2021, and issued May 03, 2023. **Patent Application**
- [P] Nikolay Chumerin, Michal Neoral, Jan Šochman, and Jiří Matas. (2020). **METHODS FOR OPTICAL FLOW ESTIMATION**. WO Patent WO2020088766A1, filed Oct 31, 2018, and issued May 07, 2020. **Patent Application**
- [T] Michal Neoral (supervisor prof. Ing. Jiří Matas, Ph.D., co-supervisor Mgr. Jan Šochman, Ph.D.) (2025). **Dense Motion Estimation in a Monocular Video**. **Doctoral Dissertation**.
- [T] Michal Neoral (supervisor Mgr. Jan Šochman, Ph.D.) (2017). **Object Scene Flow in Video Sequences**. **Master Thesis**, awarded with the Dean's Prize for outstanding master thesis.
- [T] Michal Neoral (supervisor Ing. Pavel Krsek, Ph.D.) (2014). **Extraction of Features from Moving Garment**. **Bachelor Thesis**.

PROJECTS AND INTERNSHIPS

- **Project: FunDive** Open-set and closed-set fine-grained recognition of fungal species 2026 – present
- **Project: Local CLS** Knowledge distillation for fine-grained image retrieval from frozen vision-language models 2026 – present
- **Project: CEDMO** Automated analysis of billboard video recordings for media monitoring 2026
- **Project: AISee** Video object detection and classification, incremental learning of user-defined classes 2025 – 2026
- **Project: Median** Automatic recognition of template videos in cam-glasses recordings 2025 – 2026
- **Project: Toyota Motor Europe** Computer vision research at the Toyota Research Lab, FEE CTU 2015 – 2025
- **Project: Electrolux** Identification of objects inserted into home appliances 2015
- **Project: SyRoTek** Systém pro Robotickou Tele-výuku, a robotic system for education 2014 – 2015
- **Project: CloPeMa** Clothes Perception and Manipulation, feature extraction from a moving garment 2012 – 2014
- **Toyota Motor Europe (TME), Brussels, Belgium. Internship** February 2017 – June 2017
 - Tools: C++, CUDA, Python, CAFFE2, TensorFlow
 - Developed and implementation of monocular depth estimation